

(6pts ea.) 1. Define each of the following terms:

(a) Define contrapositive.

Given a proposition $p \rightarrow q$ its contrapositive is the proposition $(\neg q) \rightarrow (\neg p)$

Alternate answer

The contrapositive of the implication $a \rightarrow b$ is the implication $(\neg b) \rightarrow (\neg a)$.

Communicates what we start with

(b) Define even.

This is here to mention which thing is even

An integer g is even if $\exists m \in \mathbb{Z}$ s.t. $g = 2m$.

Another answer → The integer k is even if there exists $g \in \mathbb{Z}$ such that $k = 2g$.

(c) Define odd.

An integer b is odd if there exists an integer p s.t. $b = 2p + 1$.

This indicates that odd applies to integers

Another

answer → Given $h \in \mathbb{Z}$ we say h is odd if $\exists a \in \mathbb{Z}$ such that $h = 2a + 1$.

(6pts ea.) 2. Define each of the following terms:

(a) Define biconditional.

Required: This tells us biconditional is a proposition

Given propositions p and q , the biconditional is the proposition that is true when p and q have matching truth value and is false when p and q have differing truth value.

This is said p if and only if q and written $p \leftrightarrow q$.

(b) Define existential quantification.

Need this. (Why?)

Given a predicate $T(x)$ and a set S , the existential quantification is the proposition that is true when $T(x)$ is true for at least one value of $x \in S$, and is false otherwise. We say "there exists $x \in S$ such that $T(x)$ " and write $\exists x \in S [T(x)]$.

(c) Define implication.

Given a proposition r and a proposition s , the implication is a proposition that is false when r is true and s is false, and in all other situations, the implication is true. The implication is written $r \rightarrow s$ and read "If r , then s ."

(6pts) 3. Show that $(p \wedge q) \rightarrow p$ is a tautology WITHOUT using truth tables.

$$\equiv (\neg(p \wedge q)) \vee p$$

$$\equiv ((\neg p) \vee (\neg q)) \vee p$$

$$\equiv ((\neg q) \vee (\neg p)) \vee p$$

$$\equiv (\neg q) \vee ((\neg p) \vee p)$$

$$\equiv (\neg q) \vee T$$

$$\equiv T$$

So $(p \wedge q) \rightarrow p$ is a tautology because $(p \wedge q) \rightarrow p$ is always true.

This sentence discusses what tautology means by specifically applying the definition to $(p \wedge q) \rightarrow p$.

(4pts) 4. Define a set that has between 2 and 1,000,000 elements.

I will give three valid answers (for the key). You should just give one.

Let A be the set of all U.S. States

Let B be the set of all UNL students.

Let $C = \{\pi, \text{Canada}, \text{pizza}\}$

There are many other possible answers.

These are two different ways to describe/define a set.

(4pts ea.)

5. For all parts on this page, let T be the set of all professional sports teams, and let $Q(x)$ be the predicate x is based in Seattle.

(a) With correct notation, state two elements of T .

One answer: Vikings, Packers $\in T$

2nd answer: Mariners $\in T$ and Lakers $\in T$

(b) State $w \in T$ in plain English.

w is a professional sports team.

This makes sense as an answer to this question since this

becomes Lakers is a professional sports team by "plugging in" for w

(c) State $\forall c \in T [Q(c)]$ in plain English.

Every professional sports team is based in Seattle.

(d) State $\exists c \in T [Q(c)]$ in plain English.

At least one professional sports team is based in Seattle.

- (5pts ea.) 6. Let $C(x)$ be the statement “ x has a cat”. Let $D(x)$ be “ x has a dog”. Let $F(x)$ be “ x has a ferret”.
Express each sentence below in terms of the predicates, quantifiers, and logical connectives using the $C(x)$ and $D(x)$ and $F(x)$ defined above. We denote by M the set of all students in MTH 225.
(a) All students in MTH 225 have a cat, a dog, or a ferret.

$$\forall s \in M [C(s) \vee D(s) \vee F(s)]$$

Also okay: $\forall z \in M [C(z) \vee D(z) \vee F(z)]$

- (b) Some student MTH 225 class has a cat, a dog, and a ferret.

$$\exists x \in M [C(x) \wedge D(x) \wedge F(x)]$$

- (c) For each of the three animals (cat, dog, ferret), there is a student in MTH 225 who has one of these animals as a pet.

$$\left(\exists a \in M [C(a)] \right) \wedge \left(\exists b \in M [D(b)] \right) \wedge \left(\exists c \in M [F(c)] \right)$$

(10^{pts}) 7. Using the hypotheses:

- H1: If Bort finds license plates, then Gus plays piano and JT uses Linux.
- H2: If Skye is a CEO, then Vish has a dog.
- H3: If JT uses Linux, then Vish does not have a dog.

prove: If Bort finds license plates, then Gus plays piano and Skye is not a CEO.

Want to show

Sentence in blue is optimal

Suppose Bort finds license plates. WTS Gus plays piano and Skye is not a CEO. Using H1, Gus plays piano and JT uses Linux due to modus ponens.

Since JT uses Linux, by H3, Vish does not have a dog due to modus ponens.

Because Vish does not have a dog, applying modus tollens to H2 tells us Skye is not a CEO.

Recalling that we earlier learned Gus plays piano, we have proved that Gus plays piano and Skye is not a CEO. $\square$

(13pts) 8. Prove: If a is even and b is odd and c is odd, then $a + bc$ is odd.

Suppose a is even and b is odd and c is odd. WTS $a+bc$ is odd.

From our three hypotheses, $\exists x, y, z \in \mathbb{Z}$ s.t. $a=2x$ and $b=2y+1$ and $c=2z+1$.

Scratch work

$$a+bc = 2x + (2y+1)(2z+1) = 2x + 4yz + 2y + 2z + 1 = 2(x + 2yz + y + z) + 1$$

Let $w = x + 2yz + y + z$. Since x, y, z are integers, $w \in \mathbb{Z}$. We note that

$$2w+1 = 2(x + 2yz + y + z) + 1 = 2x + 4yz + 2y + 2z + 1 = 2x + (2y+1)(2z+1) = a+bc,$$

which proves $\exists t \in \mathbb{Z}$ such that $a+bc = 2t+1$, which proves $a+bc$ is odd. $\square$